Mathematical Foundations of Differentiable Risk Engines: A Tensorial Formulation

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Abstract

This document presents the complete mathematical formulation underlying a differentiable seismic risk assessment engine. By encoding portfolio data as dense tensors and expressing the risk calculation as a sequence of differentiable linear algebra operations, we derive closed-form expressions for the loss function and its gradients with respect to all input variables—hazard, vulnerability, and exposure. The formulation progresses from a single-scenario deterministic setting to a fully probabilistic framework with non-uniform scenario rates, and culminates in the derivation of exact gradient expressions that enable sensitivity analysis and gradient-based optimisation without finite-difference approximations.

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1 Notation and Conventions

Throughout this document, scalars are denoted by lowercase italic letters (v_i, α), vectors by bold lowercase ($\boldsymbol{v}$, $\boldsymbol{h}$), matrices by bold uppercase ($\boldsymbol{C}$, $\boldsymbol{H}$), and index sets by standard uppercase letters (N, Q, K, M). The Kronecker delta $\delta_{a,b}$ equals 1 when $a = b$ and 0 otherwise.

Table 1 summarises the principal symbols used throughout the formulation.

Table 1: Summary of principal notation.

Symbol	Domain	Description
N	$\mathbb{Z}^+$	Number of assets in the portfolio
Q	$\mathbb{Z}^+$	Number of stochastic realizations (scenarios)
K	$\mathbb{Z}^+$	Number of vulnerability typologies
M	$\mathbb{Z}^+$	Number of discretisation points per curve
$\boldsymbol{v}$	$\mathbb{R}^N$	Exposure vector (replacement costs)
$\boldsymbol{h}$	$\mathbb{R}^N$	Intensity vector (e.g. PGA at each site)
$\boldsymbol{u}$	$\mathbb{Z}^N$	Typology index vector, $u_i \in \{1, \ldots, K\}$
$\boldsymbol{C}$	$\mathbb{R}^{K\times M}$	Vulnerability matrix (mean damage ratios)
$\boldsymbol{x}$	$\mathbb{R}^M$	Common intensity grid vector
$\boldsymbol{H}$	$\mathbb{R}^{N\times Q}$	Intensity matrix (probabilistic setting)
$\boldsymbol{\lambda}$	$\mathbb{R}^Q$	Scenario-rate vector (annual occurrence rates)
$\boldsymbol{J}$	$\mathbb{R}^{N\times Q}$	Loss matrix
$\boldsymbol{j}$	$\mathbb{Z}^N$	Grid-index vector for interpolation
$\boldsymbol{\alpha}$	$\mathbb{R}^N$	Interpolation weight vector
$\delta_{a,b}$	$\{0, 1\}$	Kronecker delta

2 Deterministic Risk Formulation

In a single-scenario analysis, we calculate the loss for N assets subjected to a given ground-motion intensity field. This section establishes the core differentiable computation upon which all subsequent developments are built.

2.1 Input Tensors

Let the portfolio consist of N assets. We define the following inputs:

- $\boldsymbol{v} \in \mathbb{R}^N$: the **exposure vector** (replacement costs).
- $\boldsymbol{h} \in \mathbb{R}^N$: the **intensity vector** (e.g. PGA at each site).
- $\boldsymbol{u} \in \mathbb{Z}^N$: the **typology index vector**, where $u_i \in \{1, \ldots, K\}$ maps asset i to one of K vulnerability curves.
- $\boldsymbol{C} \in \mathbb{R}^{K\times M}$: the **vulnerability matrix**, containing K standardised curves, each discretised at M intensity points.
- $\boldsymbol{x} \in \mathbb{R}^M$: the common **grid vector** of intensity values (abscissa of the vulnerability curves), assumed strictly increasing: $x_1 < x_2 < \cdots < x_M$.

2.2 Differentiable Interpolation

To compute the damage ratio for each asset, we must evaluate the vulnerability curve $\boldsymbol{C}_{u_i,\cdot}$ at the experienced intensity h_i. Standard lookup operations are non-differentiable; we replace them with a piecewise-linear algebraic formulation.

For each asset i, we identify the grid index j_i such that $x_{j_i} \leq h_i < x_{j_i+1}$, and compute the interpolation weight:

$$\alpha_i = \frac{h_i - x_{j_i}}{x_{j_i+1} - x_{j_i}}, \qquad \alpha_i \in [0, 1). \tag{1}$$

The Mean Damage Ratio (MDR) for asset i is then obtained by linear combination of the two bounding curve values:

$$\mathrm{MDR}_i = (1 - \alpha_i) \cdot C_{u_i, j_i} + \alpha_i \cdot C_{u_i, j_i+1}. \tag{2}$$

Risk Engineer's Interpretation

Physical Meaning: Suppose the earthquake intensity at a site falls exactly 60% of the way between 0.4 g and 0.5 g. Equation (2) computes a weighted average: 40% of the damage value at 0.4 g plus 60% of the damage value at 0.5 g. Expressing this as algebra rather than a conditional branch allows the processor to evaluate thousands of assets simultaneously via SIMD instructions.

2.3 Deterministic Loss

The total deterministic portfolio loss is the inner product of the exposure vector and the MDR vector:

$$J = \sum_{i=1}^{N} v_i \cdot \mathrm{MDR}_i = \boldsymbol{v}^\top \mathrm{MDR}. \tag{3}$$

3 Probabilistic Risk Formulation

For probabilistic risk assessment, the formulation is extended to Q stochastic realizations, each associated with an annual occurrence rate. This section generalises the deterministic computation of Section 2 to a rate-weighted ensemble, supporting non-uniform sampling of the hazard space.

3.1 Scenario-Rate Vector

We define the **scenario-rate vector**

$$\boldsymbol{\lambda} \in \mathbb{R}^Q, \qquad \lambda_q \geq 0, \tag{4}$$

where λ_q is the annual rate of occurrence (events / year) of realization q. Equally weighted realizations correspond to the special case $\lambda_q = \Lambda/Q$ for all q.

The total rate and normalised weights are

$$\Lambda = \sum_{q=1}^{Q} \lambda_q, \qquad w_q = \frac{\lambda_q}{\Lambda}, \qquad \sum_{q=1}^{Q} w_q = 1 \quad (\Lambda > 0). \tag{5}$$

3.2 Matrix Expansion of Inputs

The intensity vector $\boldsymbol{h}$ generalises to an **intensity matrix** $\boldsymbol{H} \in \mathbb{R}^{N \times Q}$, where $H_{i,q}$ is the ground-motion intensity at asset i under realization q. The static vectors $\boldsymbol{v}$ and $\boldsymbol{u}$ are broadcast along the realization axis to match dimensions (N, Q).

Correspondingly, the interpolation indices and weights become matrices:

$$j_{i,q} \in \mathbb{Z}, \qquad \alpha_{i,q} = \frac{H_{i,q} - x_{j_{i,q}}}{x_{j_{i,q}+1} - x_{j_{i,q}}} \in [0, 1). \tag{6}$$

3.3 The Loss Matrix

Applying the interpolation of Eq. (2) element-wise across assets and realizations yields the **loss matrix** $\boldsymbol{J} \in \mathbb{R}^{N \times Q}$:

$$\boldsymbol{J}_{i,q} = v_i \cdot \big[(1 - \alpha_{i,q})\, C_{u_i,\, j_{i,q}} + \alpha_{i,q}\, C_{u_i,\, j_{i,q}+1}\big]. \tag{7}$$

3.4 Risk Metrics

Annualised and distributional risk metrics are obtained by reducing the loss matrix along the realization axis using $\boldsymbol{\lambda}$:

$$\mathrm{AAL}_i = \sum_{q=1}^{Q} \lambda_q\, \boldsymbol{J}_{i,q} \qquad \text{(Average Annual Loss)}, \tag{8}$$

$$\mu_i = \sum_{q=1}^{Q} w_q\, \boldsymbol{J}_{i,q} = \frac{\mathrm{AAL}_i}{\Lambda} \qquad \text{(Mean loss per event)}, \tag{9}$$

$$\sigma_i^2 = \sum_{q=1}^{Q} w_q\, (\boldsymbol{J}_{i,q} - \mu_i)^2 \qquad \text{(Weighted variance)}. \tag{10}$$

This formulation exposes the full $(N \times Q)$ workload as dense tensor operations, naturally suited to accelerator hardware, while accommodating physically meaningful non-uniform sampling of the hazard space via $\boldsymbol{\lambda}$.

4 Gradient of Vulnerability

The tensorial architecture enables closed-form computation of the gradient of the annualised loss with respect to vulnerability model parameters. This section derives the sensitivity of the portfolio loss to perturbations of individual entries of the vulnerability matrix $\boldsymbol{C}$.

4.1 Annualised Loss Objective

Let $\bar{J}$ denote the total annualised portfolio loss:

$$\bar{J} = \sum_{q=1}^{Q} \lambda_q \sum_{i=1}^{N} v_i \big[(1 - \alpha_{i,q})\, C_{u_i,\, j_{i,q}} + \alpha_{i,q}\, C_{u_i,\, j_{i,q}+1}\big]. \tag{11}$$

4.2 Partial Differentiation

We seek $\partial \bar{J} / \partial C_{k,m}$, the sensitivity to a specific entry of the vulnerability matrix (typology k, intensity index m). Since v_i, $\alpha_{i,q}$, $H_{i,q}$, and the index pointers $j_{i,q}$ are all independent of $\boldsymbol{C}$, they are treated as constants during differentiation:

$$\frac{\partial \bar{J}}{\partial C_{k,m}} = \sum_{q=1}^{Q} \lambda_q \sum_{i=1}^{N} v_i \cdot \frac{\partial}{\partial C_{k,m}} \big[(1 - \alpha_{i,q})\, C_{u_i,\, j_{i,q}} + \alpha_{i,q}\, C_{u_i,\, j_{i,q}+1}\big]. \tag{12}$$

For a given asset i and realization q, the derivative is non-zero only when both conditions hold: (i) the asset belongs to typology k ($u_i = k$), and (ii) the interpolation segment involves node m (either as the left or right endpoint).

4.3 Unified Tensor Form

Encoding the selection logic through Kronecker deltas yields a branch-free expression:

$$\frac{\partial \bar{J}}{\partial C_{k,m}} = \sum_{q=1}^{Q} \sum_{i=1}^{N} \lambda_q \underbrace{v_i \cdot \delta_{u_i,\,k}}_{\text{asset filter}} \cdot \Big[\underbrace{(1-\alpha_{i,q})\,\delta_{j_{i,q},\,m}}_{\text{left-node contribution}} + \underbrace{\alpha_{i,q}\,\delta_{j_{i,q}+1,\,m}}_{\text{right-node contribution}} \Big]. \tag{13}$$

This derivative quantifies how the annualised portfolio loss responds to perturbations of the vulnerability curve at the point (k, m), fully accounting for non-uniform scenario rates via λ_q.

4.4 Gradient-Based Optimisation

The full gradient tensor $\nabla_{\boldsymbol{C}} \bar{J} \in \mathbb{R}^{K \times M}$ can drive iterative updates of the vulnerability parameters:

$$\boldsymbol{C}^{(t+1)} \leftarrow \boldsymbol{C}^{(t)} - \eta \cdot \nabla_{\boldsymbol{C}} \bar{J}, \tag{14}$$

where $\eta > 0$ is the step size (learning rate) and t indexes the iteration. Suitable constraints (e.g. monotonicity of damage ratios, $C_{k,m} \in [0, 1]$) may be enforced via projected gradient descent or reparametrisation.

5 Gradients of Hazard and Exposure

While vulnerability is an engineered attribute amenable to modification, hazard ($\boldsymbol{H}$) and exposure ($\boldsymbol{v}$) describe the physical event space and the portfolio state. Gradients with respect to these variables quantify local sensitivities and identify dominant contributors to portfolio risk.

The annualised loss objective is restated for convenience:

$$\bar{J} = \sum_{q=1}^{Q} \lambda_q \sum_{i=1}^{N} \boldsymbol{J}_{i,q}. \tag{15}$$

5.1 Gradient with Respect to Hazard

Applying the chain rule to the piecewise-linear interpolation of Eq. (7):

$$\frac{\partial \bar{J}}{\partial H_{i,q}} = \lambda_q\, v_i \cdot \frac{\partial \mathrm{MDR}_{i,q}}{\partial \alpha_{i,q}} \cdot \frac{\partial \alpha_{i,q}}{\partial H_{i,q}}. \tag{16}$$

Substituting $\partial \mathrm{MDR}/\partial \alpha = C_{u_i,\,j_{i,q}+1} - C_{u_i,\,j_{i,q}}$ and $\partial \alpha / \partial H = (x_{j_{i,q}+1} - x_{j_{i,q}})^{-1}$ yields:

$$\frac{\partial \bar{J}}{\partial H_{i,q}} = \lambda_q\, v_i \cdot \frac{C_{u_i,\,j_{i,q}+1} - C_{u_i,\,j_{i,q}}}{x_{j_{i,q}+1} - x_{j_{i,q}}}. \tag{17}$$

The ratio on the right-hand side is the **local slope** of the vulnerability curve at the experienced intensity. It quantifies how uncertainty or estimation error in the ground-motion intensity $H_{i,q}$ propagates into the annualised loss.

5.2 Gradient with Respect to Exposure

Since the loss function is linear in the asset values v_i, the exposure gradient follows directly:

$$\frac{\partial \bar{J}}{\partial v_i} = \sum_{q=1}^{Q} \lambda_q\, \mathrm{MDR}_{i,q}. \tag{18}$$

This is precisely the Average Annual Loss rate per unit of replacement cost for asset i. It provides the marginal annualised contribution of each asset's value to portfolio loss, accounting for the full distribution of scenarios via $\boldsymbol{\lambda}$.

Risk Engineer's Interpretation

Actionable Insight: Equation (17) reveals that assets situated on the steepest segment of their vulnerability curve are most sensitive to hazard uncertainty—this identifies where improved ground-motion modelling yields the greatest return. Equation (18) provides a natural ranking of assets by their marginal contribution to annualised risk, directly informing portfolio-level decisions such as insurance pricing or retrofit prioritisation.

6 Summary of the Gradient System

The differentiable risk formulation developed in the preceding sections yields a complete gradient system. For the annualised portfolio loss $\bar{J}$, the sensitivities with respect to each input class are collected as:

$$\nabla \bar{J} = \left[\frac{\partial \bar{J}}{\partial \boldsymbol{H}}, \frac{\partial \bar{J}}{\partial \boldsymbol{C}}, \frac{\partial \bar{J}}{\partial \boldsymbol{v}} \right], \tag{19}$$

where each component is given by:

$$\frac{\partial \bar{J}}{\partial H_{i,q}} = \lambda_q \, v_i \cdot \frac{C_{u_i,\, j_{i,q}+1} - C_{u_i,\, j_{i,q}}}{x_{j_{i,q}+1} - x_{j_{i,q}}} \qquad \text{(Hazard)}, \tag{20}$$

$$\frac{\partial \bar{J}}{\partial C_{k,m}} = \sum_{q,i} \lambda_q \, v_i \, \delta_{u_i,k} \left[(1 - \alpha_{i,q}) \, \delta_{j_{i,q},m} + \alpha_{i,q} \, \delta_{j_{i,q}+1,m} \right] \qquad \text{(Vulnerability)}, \tag{21}$$

$$\frac{\partial \bar{J}}{\partial v_i} = \sum_{q=1}^{Q} \lambda_q \, \mathrm{MDR}_{i,q} \qquad \text{(Exposure)}. \tag{22}$$

All three gradients are computed from quantities already available after the forward loss evaluation—interpolation weights $\alpha_{i,q}$, index pointers $j_{i,q}$, and the vulnerability matrix $\boldsymbol{C}$—requiring no additional simulation runs. This makes gradient computation essentially free relative to the forward pass, enabling real-time sensitivity analysis and gradient-based optimisation over the full risk model.